

VISHNU DUKNAM

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Portfolio: <https://vishnuvd45.github.io/vishnu45.github.io/>

Motivated and forward-thinking Python Developer currently pursuing a Master's in Computer Science at The University of Akron, with a strong passion for AI/ML, cloud technologies, and full-stack development. Experienced in building robust, scalable applications using Python, Java, React.js, and modern DevOps tools like Docker and Jenkins. Proven track record at Capgemini delivering impactful solutions through clean code, data analysis, and RESTful API integration. Backed by academic research in computer vision and cybersecurity, I bring a unique blend of hands-on industry experience and academic rigor. Actively seeking full-time or internship opportunities to apply my skills in developing innovative software solutions that make a real-world impact.

EDUCATION-----

Master of Science in Computer Science

August 2023-May 2025

The University of Akron.

Relevant Coursework: Database Management, Software Engineering, Data Integration, Visualization, Parallel Processing, Software Security, Human-Computer Interaction, Operating Systems, Interactive Game Design.

Bachelor of Engineering in Information Technology

May 2018 - October 2022

Anurag Group Of institutions

SKILLS-----

Programming Languages: Python, Java, JavaScript, C, C++

Java Frameworks: Spring Boot (REST API), Junit, Selenium (UI testing)

Web Technologies: React.js, Next.js, Django, Node.js, HTML5, CSS

Cloud & DevOps: AWS, GCP, Docker, Jenkins, CI/CD Pipelines

AI & Machine Learning: TensorFlow, NLP, OpenCV, Computer Vision

Databases: SQL, MySQL, PostgreSQL, MongoDB

Cybersecurity: SQL Injection Prevention, Penetration Testing

Version Control: Git, GitHub, GitLab

EXPERIENCE-----

Python Developer

Capgemini – (April 2021 – December 2023)

- Developed backend applications using Python, leveraging Pandas, NumPy, and Matplotlib for data processing and visualization.
- Performed Exploratory Data Analysis (EDA) to extract insights from datasets and improve decision-making.
- Implemented API integrations using Python to fetch and process data efficiently.
- Worked in a Linux-based development environment, managing deployments and scripting automation tasks.
- Utilized Git for version control, ensuring smooth collaboration and code management.
- Worked with IDE environments like PyCharm, VS Code, and Jupyter Notebook for efficient development.
- Followed Agile Scrum methodologies, participating in sprint planning, daily standups, and code reviews.
- Passionate about writing clean, maintainable, and well-tested code, ensuring scalability and performance.
- Gained familiarity with Big Data technologies such as PySpark for distributed data processing.

CERTIFICATIONS -----

- AWS Cloud Practitioner Essentials
- AWS Machine Learning Terminology and Process
- Google IT Support Professional Certificate

RESEARCH-----

Optimizing Driver Drowsiness Detection through Advanced Computer Vision and Image Processing Methods

- Leading a groundbreaking research project focused on enhancing driver safety by developing a highly accurate real-time drowsiness detection system. The research integrates advanced computer vision algorithms and image

processing methods to detect early signs of fatigue, with potential applications in autonomous driving and road safety technology.

Optimizing SQL Injection Detection

- Presented a paper on a hybrid approach combining machine learning (SVM) and traditional techniques for SQL injection detection. Achieved 98.68% accuracy with enhanced adaptability and reduced false positives. The proposed methodology is scalable and addresses limitations of traditional rule-based detection systems.

PROJECTS-----

Visualizing Attention in NLP Models: Technologies Used: Python

- Created an innovative open-source tool to visualize and interpret attention mechanisms in BERT, significantly improving model transparency by uncovering biases and mapping neurons to specific behaviors, used by 100+ developers worldwide.

Product Price Tracker: Technologies Used: Python, ReactJS, NextJS, MySQL

- Developed a cutting-edge data scraping tool that tracks real-time price fluctuations across multiple online platforms, enabling businesses to respond to market trends and improve decision-making efficiency by 40%.

Hostel Management System: Technologies Used: Java, HTML, CSS, JavaScript, SQL

- Engineered an intuitive student portal for room booking and management, automating processes such as ID issuance and record maintenance, streamlining administrative work and reducing manual errors by 50%.

Nice Jobs Platform: Technologies Used: MySQL, ReactJS, NextJS

- Designed and launched a dynamic online job matching platform that improved job search efficiency and recruitment processes, increasing user satisfaction by 25% within the first three months of launch.

CUDA-Based Optimizations of Bubble Sort and Merge Sort Algorithms

- Achieved significant performance boosts by optimizing Bubble Sort and Merge Sort algorithms through GPU acceleration using CUDA, enabling faster data processing for large datasets in database management and graphics rendering.

Breakout Game Application: Technologies Used: HTML, CSS, JavaScript, Canvas API

- Developed an engaging, offline-playable web-based game featuring score tracking and multiple difficulty levels, resulting in a 30% boost in user engagement through enhanced gameplay dynamics.

KEY STRENGTHS -----

- Strong time management skills, ensuring deadlines are consistently met.
- Proven teamwork and decision-making abilities in diverse environments.
- Quick adaptability to new challenges and emerging technologies.

Effective collaboration, fostering cooperation and enthusiasm in projects.
